

Taylor Pentz

Senior RAG / Vector Database Engineer

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PROFESSIONAL SUMMARY

Senior RAG / Vector Database Engineer with 10+ years delivering **enterprise SaaS, healthcare, fintech, and workflow** platforms using **SQL Server/PostgreSQL/MySQL, RAG/vector databases**, Vector databases, secure APIs, cloud services, databases, **CI/CD**, and production **observability**, creating senior-level value through modernization, **performance** tuning, reliable delivery, scalable architecture, and measurable business outcomes.

TECHNICAL SKILLS

- AI Engineering: **SQL Server/PostgreSQL/MySQL, RAG/vector databases**, Vector databases, **LLM** application design, **RAG** pipelines, prompt **workflows**, embedding generation, **vector database** retrieval, evaluation, guardrails, and production API integration.
- Application Architecture: **React, TypeScript, Node.js, Python, FastAPI, REST APIs, GraphQL**, background workers, secure identity flows, multi-tenant dashboards, and human-in-the-loop UX.
- **MLOps** & Operations: model evaluation, prompt versioning, dataset validation, **CI/CD, Docker, Kubernetes, observability**, latency monitoring, cost tracking, and rollback-ready releases.
- **SQL Server, PostgreSQL, MySQL, MongoDB, Redis**, indexing, query tuning, schema design, migration planning, transaction handling, data validation, and **performance** profiling.
- **AWS, Azure, GCP, Docker, Kubernetes, GitHub Actions, Azure DevOps, Jenkins, Terraform** basics, environment configuration, blue-green releases, and deployment gates.
- **xUnit, NUnit, Jest, Cypress, Playwright, Postman**, integration testing, contract testing, regression testing, code reviews, **SonarQube**, and secure coding practices.
- **Application Insights, CloudWatch, ELK Stack, Prometheus, Grafana, OpenTelemetry**, structured logging, alerting, production incident review, root-cause analysis, and SLO-focused engineering.
- **Senior** Practices: AI product discovery, architecture reviews, **security** reviews, mentoring, documentation, experimentation discipline, stakeholder communication, and responsible delivery.

PROFESSIONAL EXPERIENCE

SumatoSoft

Senior Software Engineer

Boston, MA (Remote) | Apr 2023 - Dec 2025

- Led delivery of **enterprise SaaS and workflow** modernization features with **RAG, Vector Database**, AI-enabled applications, **RAG workflows**, model integration, evaluation, APIs, and production **observability**, secure data access, reusable architecture, and cloud services for distributed business users.
- Designed role-focused feature modules, service boundaries, and shared engineering patterns that improved release consistency and increased feature delivery speed by **28%** across product teams.
- Modernized legacy application areas by upgrading framework dependencies, resolving package conflicts, fixing build pipeline failures, and replacing brittle patterns with maintainable senior-level architecture.
- Resolved production **performance** problems caused by inefficient data access, oversized payloads, missing indexes, slow rendering paths, and noisy integrations, reducing high-latency **workflows** by **34%**.
- Implemented **OAuth2/OIDC** authentication, **JWT** validation, **RBAC** authorization, audit logging, tenant-aware data access, and secure configuration practices for administrative and customer-facing platforms.
- Refactored tightly coupled service logic into **Clean Architecture, CQRS**-style handlers, domain services, and testable modules that made complex business rules easier to extend and review.
- Built new capabilities for case management, reporting dashboards, notification preferences, file **workflows**, user administration, and operational approvals using **RAG, Vector Database** and pragmatic engineering standards.
- Improved production **reliability** by adding structured logging, health checks, retry policies, circuit breakers, dashboards, and incident-focused diagnostics around APIs, jobs, and external services.
- Enhanced **CI/CD** pipelines with automated tests, environment controls, **Docker**-based builds, deployment gates, and rollback-friendly release practices to reduce failed deployments and improve confidence.

Hexaview Technologies Inc.

Software Engineer

New York, NY | Jul 2018 - Mar 2023

- Developed customer-facing **fintech** and **enterprise** applications with **Python**, API-backed analytics tools, search **workflows**, data-driven features, secure service integration, and production-ready data services, secure APIs, relational databases, caching, and automated delivery pipelines for high-value business **workflows**.
- Migrated older modules into modern service boundaries, reducing duplicated logic and improving deployment flexibility across onboarding, payment, account, document, and reporting **workflows**.
- Implemented reusable screens, service layers, background processes, and integration **workflows** using **Python**, analytics APIs, search **workflows**, and data services, practical testing patterns, and production-oriented review standards.

- Reduced recurring production defects by **31%** by improving validation rules, standardizing API error handling, adding regression coverage, and mapping exceptions into clearer support diagnostics.
- Solved timeout issues in payment, reporting, and customer-data **workflows** by optimizing SQL queries, cache usage, pagination, background processing, and transaction boundaries for large datasets.
- Supported modernization from older framework versions and legacy UI patterns into component-based, API-first, and typed service approaches that were easier to maintain and onboard.
- Built secure integrations with payment, identity, document, messaging, and reporting platforms using **OAuth2**, webhooks, retry queues, idempotency controls, and detailed audit trails.
- Strengthened engineering quality through automated tests, **Postman** collections, peer reviews, **SonarQube** checks, and collaborative Agile planning with product, QA, DevOps, and client stakeholders.

SimbirSoft Company

Software Developer

Boston, MA | Mar 2015 - Jun 2018

- Built **enterprise** business applications using **C#**, **JavaScript**, **SQL Server**, API-backed web **workflows**, layered architecture, and **enterprise** debugging practices, relational databases, and IIS-hosted services for operations, reporting, approval, and customer-record **workflows**.
- Developed backend services and UI **workflows** for order processing, customer records, scheduled jobs, export features, internal dashboards, and transaction-safe operational updates.
- Improved page and **workflow** responsiveness by **26%** by reducing unnecessary round trips, optimizing SQL execution plans, trimming payloads, and replacing heavy server-rendered patterns.
- Resolved production issues involving null reference errors, duplicate form submissions, broken session handling, slow exports, and inconsistent validation by improving logging and defensive checks.
- Participated in legacy modernization by moving selected monolithic screens toward service-backed APIs, reusable components, Bootstrap-based layouts, and cleaner separation of responsibilities.
- Implemented access checks, dashboard widgets, scheduled notifications, import/export routines, and reporting improvements that helped business users complete tasks with fewer manual spreadsheets.
- Created unit, integration, API, and SQL test data checks that improved refactoring confidence, reduced regression risk, and supported predictable monthly release cycles.

Microsoft

Software Developer Intern

Cambridge, MA | Aug 2014 - Feb 2015

- Supported internal engineering tools with **C#**, **JavaScript**, **SQL Server**, API-backed web **workflows**, layered architecture, and **enterprise** debugging practices, **SQL Server** data flows, **JavaScript**-based administrative screens, and structured **enterprise** development practices in a large software environment.
- Built small backend features for internal dashboards, including data-entry endpoints, validation flows, reporting views, and service-layer logic used by engineering and operations teams.
- Fixed UI defects involving inconsistent form validation, browser rendering issues, broken table sorting, and missing client-side error messages across administrative **workflow** screens.
- Assisted with SQL query cleanup, stored procedure updates, and data-mapping corrections to improve report accuracy and reduce support noise from internal business users.
- Added logging, defensive null checks, and clearer exception messages to help senior engineers diagnose recurring staging issues during release validation and regression testing.
- Created test cases, manual regression scripts, and API checks while gaining practical experience with Agile ceremonies, pull requests, code reviews, and **enterprise** coding standards.

SELECTED PROJECTS

- AI-Assisted Enterprise Knowledge Platform: Designed a secure **RAG**-enabled knowledge **workflow** using **SQL Server/PostgreSQL/MySQL**, **RAG/vector databases**, Vector databases, vector retrieval, role-based access, API orchestration, evaluation checks, and **observability** to improve support response consistency and reduce repeated research time.
- Workflow Automation and Reporting Platform: Built a senior-level modernization project using **SQL Server/PostgreSQL/MySQL**, **RAG/vector databases**, Vector databases, secure APIs, relational data stores, **CI/CD**, dashboards, and structured logging to support tenant-aware business **workflows** and reliable executive reporting.

EDUCATION

University of North Carolina at Chapel Hill - Bachelor's Degree in Computer Science (2010 - 2014)